ARANYA SAHA

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RESEARCH INTERESTS

The goal of my research is to develop the foundations of efficient, reliable, and interpretable machine learning and translate them into practical AI systems. I am particularly interested in advancing large language models through novel perspectives from optimization, information theory, and reinforcement learning, with research interests spanning efficient LLMs, mechanistic interpretability, machine unlearning, and AI for Science.

EDUCATION

University of Maryland, College Park (UMD) Aug 2026 – Present
PhD in Electrical and Computer Engineering
Research Advisor: Prof. Sanghamitra Dutta

Bangladesh University of Engineering and Technology (BUET) Feb 2020 – Mar 2025
BSc in Electrical and Electronic Engineering
Research Advisor: Prof. Mohammad Ariful Haque

RELEVANT COURSEWORK

Graduate	Random Processes, Advanced Digital Signal Processing, Information Theory, Statistical Pattern Recognition
Undergraduate	Artificial Intelligence and Machine Learning, Digital Signal Processing, Random Signals and Processes, Probability and Statistics, Signals and Systems, etc.

PUBLICATIONS

** denotes equal contribution*

1. Ismam Nur Swapnil, **Aranya Saha**, Tasneea Zahra, Tanvir Ahmed Khan, Mohammad Ariful Haque, Ser-Nam Lim, "Skip What You Can Predict: Predictive Repositioning for Policy Optimization for Efficient LLM Training" – Preprint *(Under review at ICLR 2027)*
2. Vivek Chowdhury*, **Aranya Saha***, Archisman Sarker*, Mohsinur Rahman Adnan, "Deep Learning-Based Automated Parameter Extraction for Sub-60 mV/dec HfSiO-Based Negative-Capacitance Field-Effect Transistors" *(Under review at APL Machine Learning Journal)*
3. Ismam Nur Swapnil*, **Aranya Saha***, Tanvir Ahmed Khan*, Mohammad Ariful Haque, "GRPO++: Enhancing Dermatological Reasoning under Low Resource Settings" – Preprint *(Under review at IEEE Journal of Biomedical and Health Informatics)*
4. **Aranya Saha***, Tanvir Ahmed Khan*, Ismam Nur Swapnil*, Mohammad Ariful Haque, "CLARIFY: A Specialist–Generalist Framework for Accurate and Lightweight Dermatological Visual Question Answering" – Preprint
5. Tanvir Ahmed Khan, **Aranya Saha**, Ismam Nur Swapnil, Mohammad Ariful Haque, "Compression Strategies for Efficient Multimodal LLMs in Medical Contexts" – Preprint
6. Shadman Sobhan, **Aranya Saha**, Tanvir Ahmed Khan, Abduz Zami, "Skin Cancer Classification Using Pre-trained CNNs: A Transfer Learning Approach Addressing Imbalanced Data Challenges," International Conference on Next-Generation Computing, IoT and Machine Learning (2025) – Link
7. Shadman Sobhan, Abduz Zami, Mohiuddin Ahmed, Tanvir Mahtab Zihan, Tanvir Ahmed Khan, **Aranya Saha**, "A Multi-Stage Deep Learning Approach to Tuberculosis Detection with Explainable Insights," International Conference on Next-Generation Computing, IoT and Machine Learning (2025) – Link

SELECTED PROJECTS

1. **Efficient Frame Selection for Long Egocentric Video Understanding.** Developed an efficient frame selection approach for long-form egocentric video understanding. – GitHub
2. **EchoLens: Multimodal Conversational AI Engine.** Developed a multimodal conversational AI system integrating vision and language models for interactive understanding and response generation. – GitHub
3. **Simple MedQA-GPT: GPT Tailored for Medical Q&A.** Developed a GPT-based system for answering medical questions with domain-specific prompting and conversational interaction. – GitHub

4. **Autonomous Delivery Drone for Remote and Inaccessible Areas.** Designed an autonomous drone-based delivery system for transporting essential supplies to remote and inaccessible locations. – Report
5. **IoT-Based Patient Health Monitoring System.** Developed an IoT-enabled system for real-time monitoring of patient health parameters. – GitHub Report
6. **Designing a 4-bit ALU with Memory.** Designed and implemented a 4-bit arithmetic logic unit integrated with memory for basic computational operations. – Report
7. **Multi-Functional CNC: Pen Plotter, Cutter, and Engraver.** Designed and implemented a multifunctional CNC machine capable of automated plotting, cutting, and engraving. – Report
8. **MATLAB-Based Fingerprint Recognition System.** Developed a fingerprint recognition system using MATLAB for biometric identification and matching. – GitHub Report

EXPERIENCE

University of Maryland, College Park Aug 2026 – Present
Graduate Research Assistant (FORMAL Group) *MD, USA*

Advisor: Prof. Sanghamitra Dutta

Advanced Chemical Industries PLC Apr 2025 – Aug 2026
Machine Learning Engineer (MIS Team) *Dhaka, Bangladesh*

Projects: LLM-Powered Automated CV Evaluation; LLM-Powered Tabular Data Analysis; Bengali OCR for Printed and Handwritten Text; Computer Vision-Based Fraud Detection for Self-Checkout.

Bangladesh University of Engineering and Technology Nov 2025 – Apr 2026
Part-time Lecturer (Electrical and Electronic Engineering) *Dhaka, Bangladesh*

Instructed Digital Signal Processing Sessional, Wireless Communication Laboratory, Robotics and Automation Laboratory, and Basic Electrical Circuits Laboratory courses.

TECHNICAL SKILLS

Programming	Python, C/C++, MATLAB
ML/DL	PyTorch, TensorFlow, CNNs, VLMs, LLMs
Data/Tools	NumPy, Pandas, Git, Docker, FastAPI, LaTeX
Hardware	Microcontrollers, IoT Devices, Sensors

LEADERSHIP EXPERIENCE

Association for Computing Machinery (ACM) Apr 2024 – Feb 2025
Student Executive (SIGCOMM) *Remote*

Served as the first Student Executive of ACM SIGCOMM, working alongside Prof. Matthew Caesar (UIUC), Chair of ACM SIGCOMM; developed its official website under his mentorship and co-established a paper reading group.

HONORS AND AWARDS

1. Dean's Fellowship (ECE) – University of Maryland, College Park (Fall 2026, Spring 2027).
2. 1st Runner-Up, Poster Competition (AI) – BEAR Summit 2025, Bangladesh.
3. University Merit Scholarship – BUET, awarded for academic performance (4 semesters).
4. Dean's List Award – BUET, awarded for academic performance (Years 1–2).
5. Ranked 29th among 10,000+ candidates in the BUET Undergraduate Admission Test (2019).
6. Ranked 31st (Male) in the Dhaka Board HSC Examination, Bangladesh; recipient of the Talent Pool Scholarship.
7. Perfect Attendance Certificate (11th and 12th Standard) – Notre Dame College, Dhaka, Bangladesh.

REFERENCES

Prof. Sanghamitra Dutta
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